

Experiment No: 03

Name of the Experiment: Zener Diode applications.

Objective:

Study of the Zener Diode applications.

Theory:

The diodes we have studied before do not operate in the breakdown region because this may damage them. A Zener diode is different; it is a silicon diode that the manufacturer has optimized for operation in the breakdown region. It is used to build voltage regulator circuits that circuits that hold the load voltage almost constant despite large change in line voltage and load resistance. The symbol of Zener diode shows in figure 4.1.

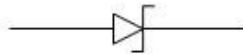


Figure 3.1 : Symbol of Zener Diode.

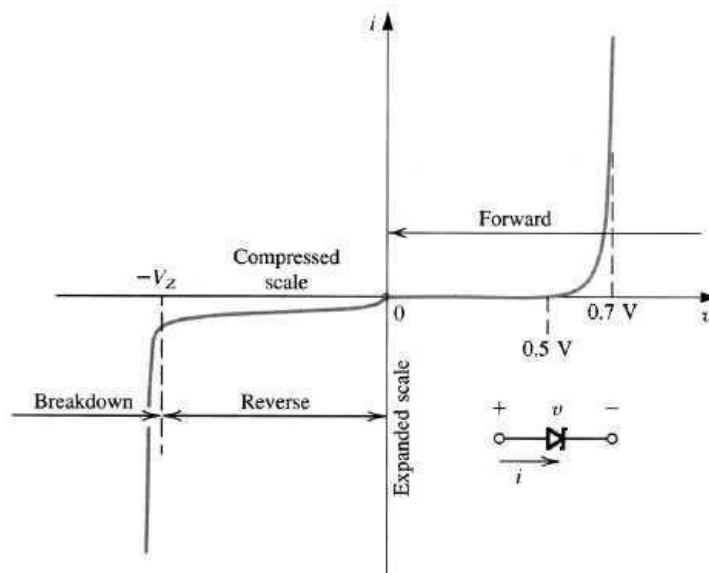


Figure 3.2 : I - V Characteristics of Zener Diode.

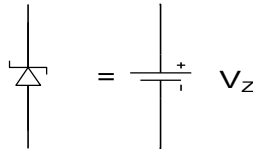
The Zener diode may have a breakdown voltage from about 2 to 200 volts. These diodes can operate in any of three regions – forward, leakage and breakdown. Figure 4.2 shows the I-V characteristics curve of Zener diode.

- In the forward region it works as an ordinary diode.
- In the leakage region (between zero and breakdown) it has only a small reverse saturation current.

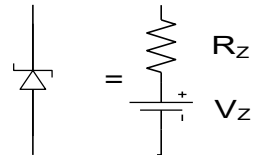
- In the breakdown it has a sharp knee, followed by an almost vertical increase in current without changing the voltage.
- The voltage is almost constant, approximately equal to V_Z over most of the breakdown region.

Equivalent circuits of Zener Diode : Two approximation are used for Zener Diode equivalent circuit.

First Approximation : As the voltage remains constant across the Zener diode though the current changes through it, it is considered as a constant voltage source according to the first approximation.



Second Approximation : A Zener resistance is in series with the ideal voltage source is approximated.



Equipment and Components:

Serial no.	Component Details	Specification	Quantity
1.	Zener diode	5 volts	1 piece
2.	Resistor	220Ω, 470Ω, 1KΩ	1 piece each
3.	POT	10KΩ	1 unit
4.	Trainer Board		1 unit
5.	DC Power Supply		1 unit
6.	Digital Multimeter		1 unit
7.	Chords and wire		as required

Experimental Setup:

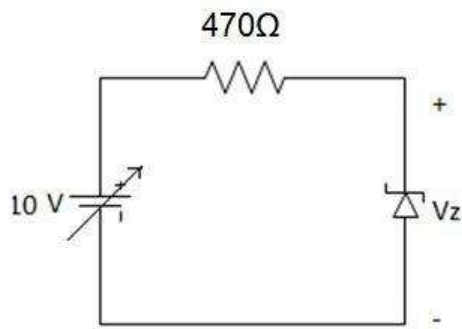


Figure 3.3: Reverse Biased Zener Characteristics

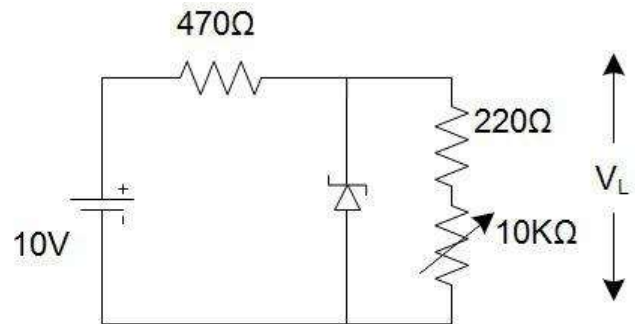


Figure 3.4: Load Regulation

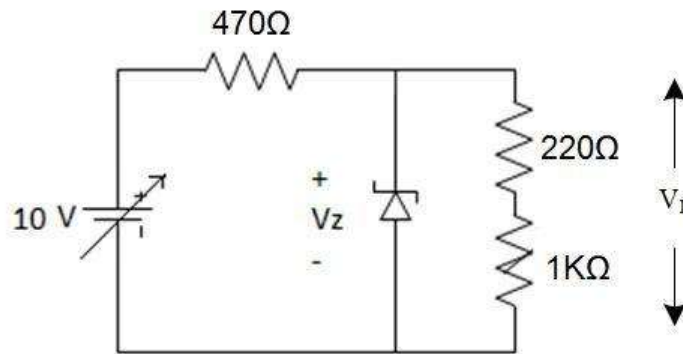


Figure 3.5: Line Regulation

Procedure:

1. Connect the circuit as shown in the figure 3.3
2. Vary the supply voltage from zero volt, complete the Table 3.1.
3. Connect the circuit as shown in the figure 3.4
4. Keep the POT at maximum position and power up the circuit.
5. Gradually decrease the POT resistance and complete the Table 3.2.
6. Replace the POT with $1K\Omega$ resistance, vary the supply voltage and take reading for Table 3.3.

Data Collection:

Signature of instructor:

Experiment: 3,
Performed by Group# _____

Theoretical	Practical
470 Ω	
220 Ω	
1K Ω	

Table 3.1: Data for I - V characteristics.

V (volts)	V_R (volts)	V_Z (volts)	$I_Z = V_R / R$ (ma)
0.1			
0.3			
0.5			
0.3			
1.0			
2.0			
3.0			
4.0			
5.0			
6.0			
7.0			
8.0			
9.0			
10.0			

Table 3.2: Data for Load Regulation

POT R (k Ω)	V ₂₂₀ (mV)	V _L (volts)	I _L (Amp)= V ₂₂₀ /220

Table 3.3: Data for Line Regulation.

V (volts)	V _L (volts)
1.0	
3.0	
6.0	
8.0	
9.0	
10.0	
11.0	
12.0	

Report:

1. Plot I_Z vs V_Z characteristics of Zener diode. Determine the Zener breakdown voltage from the plot.
2. Plot I_L vs V_L for the data table 4.2. Scale [x-axis: 0.1V/DIV, y-axis: any suitable range]. Find the Load regulation from the graph.
3. Plot V_L vs V for the data table 4.3. Find the line regulation from graph.

Objective: study of Zener Diode applications.

Theory:

A Zener diode is a specially designed silicon PN Junction diode that is intended to operate safely in reverse breakdown region. Unlike an ordinary diode, which may be permanently damaged if reverse breakdown occurs, a Zener diode is manufactured to withstand break-down while maintaining stable electrical performance. Because of this property, it is widely used in voltage regulation, voltage reference circuit, and over-voltage protection systems.



Fig 1: symbol of zener diode.

The operation of a Zener diode can be understood from its current-voltage ($I-V$) characteristics, shown in the figure 2. The device operates in three distinct regions: forward bias, reverse leakage, and reverse breakdown.

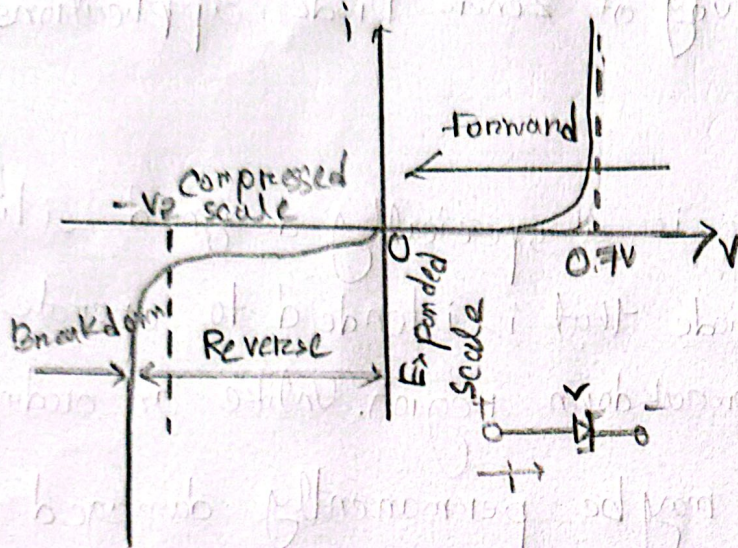


Fig 2: I-V characteristics of Zener diode

* Forward Bias Region:

When the Zener diode is forward biased, it behaves like a conventional silicon diode. The current remains very small until the forward voltage reaches approximately 0.7V. Beyond this value, the forward ~~current~~ current increases rapidly with only a small increase in voltage.

* Reverse Breakdown Leakage Region:

When reverse voltage reaches the Zener breakdown voltage V_Z , break down occurs. At this point, the reverse current increases sharply, while the voltage across the diode remains nearly constant. This constant-voltage characteristic makes the Zener diode an ideal device for voltage regulation. The breakdown voltage

of commercially available Zener diodes typically ranges from 2V to 200V, depending on the device specification.

* Reverse Leakage Region:

When reverse voltage is applied ^{below} but remain ^{below} the breakdown voltage, only a very small reverse leakage current flows through the diode. In this region the diode behaves almost like an open circuit.

* Equivalent Circuit of a Zener diode:

To simplify circuit analysis, two ~~pro~~ practical equivalent models are commonly used.

* First Approximation:

In the first approximation, the voltage across the Zener diode is assumed to remain perfectly constant after breakdown. Therefore the diode is ~~is~~ represented by an ideal constant voltage source, equal to V_Z .

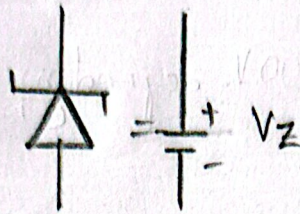


Fig. 1:

This

this model is suitable for basic circuit analysis where high accuracy is not required.

~~A~~ Second Approximation:

In practice, the voltage across a Zener diode changes slightly as the current varies. To

~~accurate~~ account for this effect, the second approximation ~~represents~~ the diode as an ideal Zener diode voltage source (V_Z) in series with a small dynamic resistance (R_Z). This model provides a more accurate representation of the practical behavior of the Zener diode, especially when analyzing voltage regulation under varying load conditions.

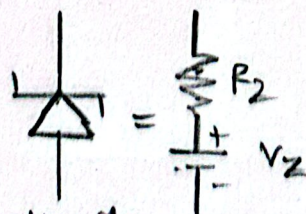


Fig. 2:

Application As a Voltage Regulator:

A Zener diode is commonly connected in reverse biased across the load with a series resistor connected to the supply. The series resistor limits the current through the diode, while Zener diode maintains an almost constant output voltage equal to V_Z .

If the input voltage increases, the Zener current increases, but the output voltage remains nearly constant. Similarly, when the load current changes, the Zener current adjusts automatically to maintain a stable output voltage. This property makes the Zener diode an effective and reliable voltage regulator in electronic circuits.

The Zener diode is a reverse-biased semiconductor device specifically designed to operate in the breakdown region. Its ability to maintain an almost constant voltage over a wide range of current makes it essential in voltage regulation and reference

applications. The first and second approximation models simplify circuit analysis while providing practical insight into the operating characteristics of the device.

Equipment and Components:

Serial No	Component Details	Specification	Quantity
1.	Zener diode	5 volts	1 piece
2.	Resistor	220 Ω , 470 Ω , 1k Ω	1 piece each
3.	Pot	10 k Ω	1 unit
4.	Trainer board		1 unit
5.	DC power supply		1 unit
6.	Digital Multimeter		1 unit
7.	Chords and wire		1 unit

Circuit Diagrams:

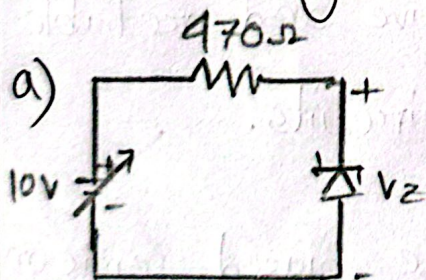


Fig: Reverse Biased Zener Characteristics

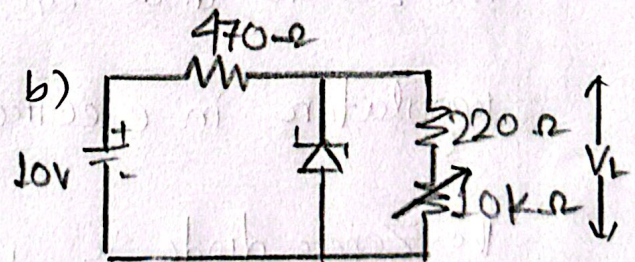
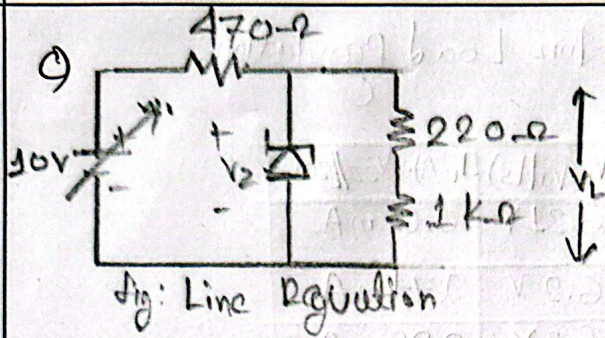


Fig: Load Regulation



Data collection:

Theoretical	Practical
470 Ω	470 Ω
220 Ω	219.8 Ω
1k Ω	0.995 k Ω

Table 4.1 : Data for I-V characteristics:

V(volts)	V _R (volts)	V _Z (volts)	I _Z = V _R /R (mA)
0.1	0	74.3mV	0
0.3	0	291mV	0
0.5	0	0.40V	0
0.8	0	0.846V	0
1.0	0	1.01V	0
2.0	0	2.04V	0
3.0	0	3.02V	0
4.0	0	4.03V	0
5.0	0	4.99V	0
6.0	0.8mV	6.07V	0.002mA
7.0	0.85mV	6.16V	1.81mA
8.0	1.84V	6.17V	3.01mA
9.0	2.17V	6.16V	4.62mA
10.0	3.84V	6.2V	8.17mA

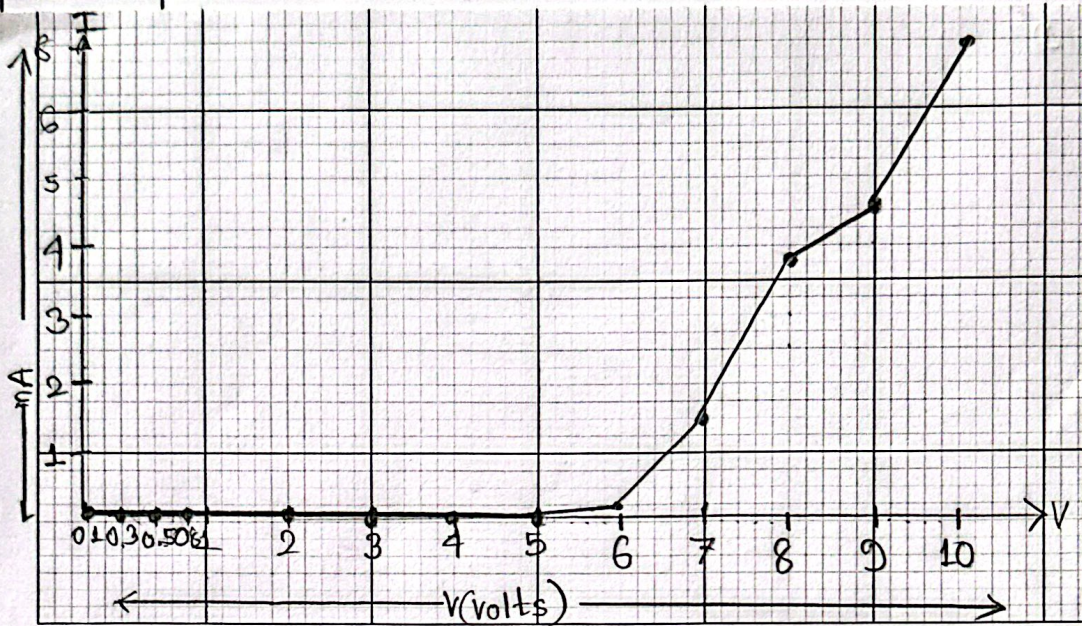
Data table 4.2: Data for Load Regulation:

Pot R (k Ω)	V_{220} (mV)	V_L (volts)	I_L (A) = $V_{220}/220$
10	136.8 mV	6.21 V	0.62 mA
8	162.7 mV	6.2 V	0.74 mA
6	214.2 mV	6.2 V	0.98 mA
5	259.4 mV	6.2 V	1.18 mA
4	312.5 mV	6.2 V	1.42 mA
2	0.603 V	6.19 V	2.74 mA
1	1.06 V	6.18 V	4.82 mA

Data table 4.3: Data for Line Regulation

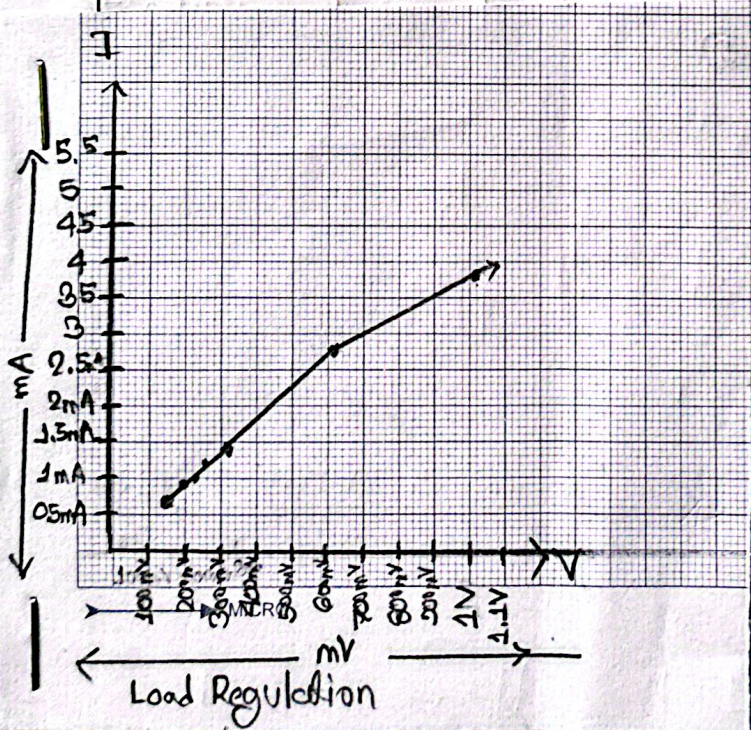
V_C (volts)	V_L (volts)
1.0	0.75 V
3.0	2.183 V
6.0	4.37 V
8.0	5.83 V
9.0	6.14 V
10.0	6.16 V
11.0	6.17 V
12.0	6.20 V

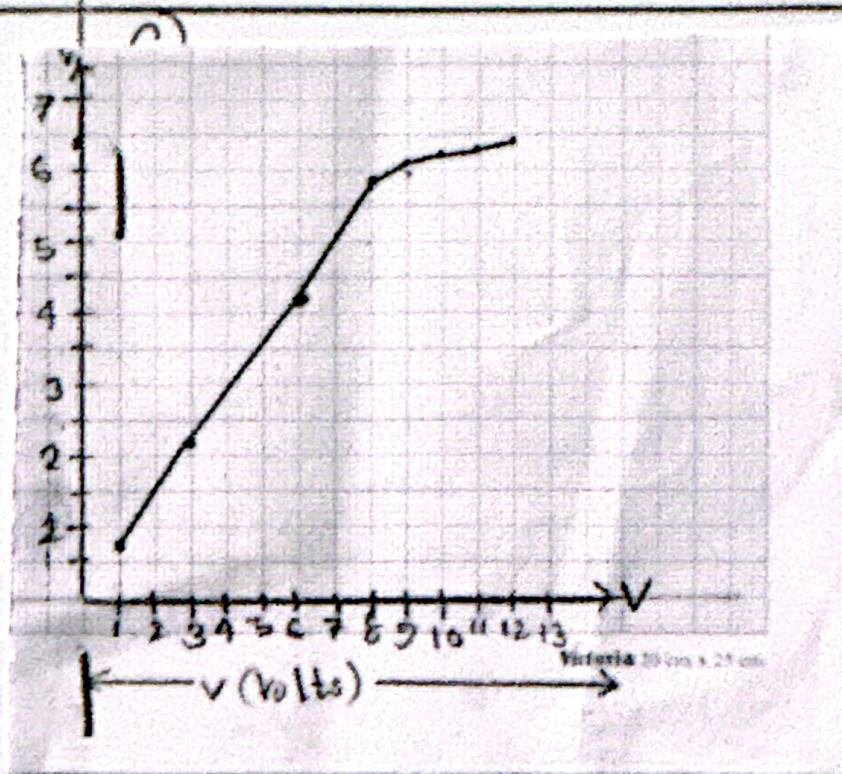
Graph:



Reverse Biased Zener characteristics

b)





Result Analysis:

The Objective of this experiment was to investigate the I-V characteristics of Zener diode and to verify its performance as voltage regulator under different load and supply conditions. Three sets of experimental data were obtained: The reverse I-V characteristics of the Zener diode, load regulation and the line regulation.

From data table 4.1, the voltage across the Zener diode increased with the applied supply voltage until it reached approximately 6.1-6.2V. Beyond this

point, the Zener diode voltage remained almost constant while the Zener current increased from 0mA to approximately 8.17mA. The plotted $I_Z - V_Z$ characteristic exhibited a distinct knee at the breakdown region, confirming that the diode entered reverse breakdown at approximately 6.2V. This behavior agrees with the theoretical characteristic of a Zener diode, where a large increase in current produces only a negligible change in voltage after breakdown.

In load regulation experiment (table 4.2), the load resistance was gradually decreased, causing the load current to increase from 0.62mA to 4.82mA. Despite this significant change in load current, the load voltage remained nearly constant between 6.18V and 6.24V. The $I_L - V_L$ graph therefore demonstrated that the Zener diode effectively maintained a stable output voltage under varying load conditions. This verifies the voltage regulation capability of Zener diode.

For the regulator experiment (table 4.3), the input supply voltage was varied from 1V to 12V. Initially, the load voltage increased with the supply voltage because the Zener diode had not yet reached its breakdown region. Once the input voltage exceeded approximately 9V, the output voltage stabilized around 6.16V-6.20V, even though the input voltage continued to increase. The V_L - V graph clearly illustrates that after breakdown, the Zener diode maintained an almost constant output voltage despite changes in the input supply voltage.

Overall, the experimental results closely matched the theoretical behavior of a Zener diode. The measured breakdown voltage was approximately 6.2V, and both the load regulation and line regulation experiments confirmed that Zener diode successfully maintained a nearly constant output voltage under varying operating conditions, therefore the objects of the experiment were successfully achieved.

Questions and Answers:

- 1) already done in graph section (a)
- 2) already done in graph section (b)
- 3) already done in graph section (c)

Discussion:

This experiment helped us understand the practical operation of a Zener diode and its importance in voltage regulation circuits. Although the theoretical concepts were familiar from lectures, performing the experiment provided a clear understanding of how a Zener diode behaves in different operating conditions and how it maintains a stable output voltage in practical applications. It also improved our confidence in circuit construction, instrument handling, and accurate data collection.

The experimental data and the corresponding graphs were consistent with the expected theoretical behaviors of the Zener diode. Minor differences between

the theoretical and practical values were observed, which are common in laboratory experiments. These variations can be attributed to the tolerance of electronic components, the internal resistance of measuring instruments, and small fluctuation in the DC power supply. However, these differences were insignificant and did not affect the overall outcome of the experiment.

During the experiment, we did not encounter any technical or equipment-related problems. All the instruments operated properly, and the circuit functioned correctly on the first attempt. Since the experimental setup was completed smoothly and the readings were stable, we were able to finish the experiment within a short period of time while maintaining accuracy in our measurements.

Overall, this experiment strengthened our understanding of the practical application of Zener diodes and highlighted the importance of careful circuit assembly, precise measurement, and systematic observation in obtaining reliable experiment results. The knowledge gained from this experiment will be valuable for future studies involving voltage regulators and other semiconductor-based electronic circuits.

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Procedure:

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6. Replace the POT with $1K\Omega$ resistance, vary the supply voltage and take reading for Table 3.3.

Data Collection:

Signature of instructor:

Experiment: 3,
Performed by Group# 11

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Theoretical	Practical
470 Ω	470 Ω
220 Ω	216Ω 219.8 Ω
1K Ω	0.995K Ω

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0.5	0	0.49V	0
0.8	0	0.846V	0
1.0	0	1.01V	0
2.0	0	2.04V	0
3.0	0	3.02V	0
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Table 3.2: Data for Load Regulation

POT R (k Ω)	V ₂₂₀ (mV)	V _L (volts)	I _L (Amp)= V ₂₂₀ /220
10	136.8mV	6.21V	0.62mA
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Table 3.3: Data for Line Regulation.

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1. Plot I_Z vs V_Z characteristics of Zener diode. Determine the Zener breakdown voltage from the plot.
2. Plot I_L vs V_L for the data table 4.2. Scale [x-axis: 0.1V/DIV, y-axis: any suitable range]. Find the Load regulation from the graph.
3. Plot V_L vs V for the data table 4.3. Find the line regulation from graph.